

RESEARCH ARTICLE



Threshold analysis of mortality outcomes in the collaborative ocular melanoma study (COMS)

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ABSTRACT

Aim: The collaborative ocular melanoma study (COMS) reported similar survival between plaque brachytherapy (a type of interventional radiotherapy) and enucleation for medium-sized choroidal melanomas. We aimed to quantify the mortality differences required to achieve statistical significance and assess the robustness of these thresholds.

Methods: We reanalyzed 12-year mortality data from COMS using threshold analysis to determine how many additional or fewer deaths in either treatment arm would shift the lower bound of the 95% confidence interval (CI) for the risk ratio above 1.

Results: At 12 years, there were 105 melanoma-related deaths in the brachytherapy arm and 98 in the enucleation arm (risk ratio 1.08, 95% CI 0.82–1.42). Achieving statistical significance would have required 31 additional deaths in the brachytherapy arm or 23 fewer in the enucleation arm; conversely, favoring brachytherapy would have necessitated 34 additional deaths in the enucleation arm or 27 fewer in the brachytherapy arm. These thresholds remained consistent across power levels and sensitivity scenarios.

Conclusions: The COMS trial did not reveal a clinically meaningful survival difference. Our findings underscore the substantial mortality shifts required for statistical significance and highlight the challenges in detecting modest treatment effects in uveal melanoma trials.

ARTICLE HIGHLIGHTS

- The COMS trial reported comparable 12-year survival between brachytherapy and enucleation for medium-sized choroidal melanomas.
- At 12 years, the observed risk ratio was 1.08 (95% CI 0.82–1.42), based on 105 deaths in the brachytherapy arm and 98 in the enucleation arm.
- Our threshold analysis quantifies the mortality change needed in either arm to shift the 95% CI of the risk ratio beyond 1.
- Achieving statistical significance in favor of enucleation would have required 31 additional deaths in the brachytherapy arm or 23 fewer deaths in the enucleation arm.
- Conversely, favoring brachytherapy would have required 34 additional deaths in the enucleation arm or 27 fewer deaths in the brachytherapy arm.
- The required changes in mortality remain consistent across various statistical power levels.
- Sensitivity analyses using uniform adjustments to both arms (–40%, –20%, +20%, +40%) confirmed that the thresholds for significance are substantial.
- The analysis illustrates the challenge of detecting modest treatment effects given the high baseline mortality associated with metastatic uveal melanoma.
- The insights from this study provide guidance for the design and sample size considerations for future randomized trials in uveal melanoma.

PLAIN LANGUAGE SUMMARY

Uveal melanoma is the most common eye cancer in adults, and it is deadly if it spreads to other body parts. The collaborative ocular melanoma study (COMS) compared two treatments for medium-sized tumors: brachytherapy (a type of radiation therapy) and enucleation (surgical removal of the eye). The original study found no significant difference in survival between the two treatments after 12 years.

In our research, we reexamined the COMS data to determine how many more deaths would have needed to occur in one treatment group or fewer in the other to show a significant survival difference. We found that a large number of additional deaths would be required to

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make one treatment appear better than the other. This suggests that the COMS trial was not able to detect small but potentially important differences in survival between the treatments. Our findings highlight the difficulties in conducting studies for uveal melanoma and suggest that future research should use larger patient groups and more advanced methods to better understand the effectiveness of different treatments.

1. Introduction

The management of uveal melanoma, the most common primary intraocular cancer in adults, continues to pose a significant challenge. Thirty years post-diagnosis, mortality from metastatic uveal melanoma approaches 50%, with limited improvements in outcomes over recent decades [1,2].

The collaborative ocular melanoma study (COMS) stands as one of the few randomized clinical trials conducted for the treatment of primary choroidal melanoma and remains the largest study in ocular oncology. With 43 participating centers, over 2000 patients, 28 numbered publications, and numerous additional publications stemming from its findings, the knowledge generated by the COMS continues to influence everyday clinical practice across the globe [3–5]. The trial's goal was to detect or rule out risk ratios for mortality (overall and with melanoma metastasis) of 1.25 and 0.75, and the investigators achieved those goals. In 1317 patients with medium-sized choroidal melanoma (2.5–10mm in apical height and ≤ 16 mm in basal diameter), the mortality from histopathologically confirmed melanoma metastasis was no worse following brachytherapy with iodine-125 compared to enucleation [6,7]. The 12-year mortality rate from melanoma metastasis was 21% in the brachytherapy arm and 17% in the enucleation arm, with an unadjusted risk ratio for death with histopathologically confirmed melanoma metastasis of 1.07 (95% confidence interval [CI] 0.81 to 1.41) for patients treated with brachytherapy through 12 years.

However, preexisting metastatic disease in some patients may have influenced mortality outcomes, potentially confounding the comparison between treatment modalities. It is estimated that micrometastases can disseminate from the primary tumor several years before treatment, and such micrometastases have been identified in patients without clinically detectable metastatic disease [8–10]. To address this, our study expands upon the original COMS data to estimate the number of additional or reduced deaths from histopathologically confirmed melanoma metastasis required to achieve a statistically significant difference between the two treatment groups. By conducting a threshold analysis, we aim to determine the magnitude of effect necessary for the trial to demonstrate the superiority of one treatment over the other.

This quantitative assessment offers valuable insights into the COMS trial's statistical power limitations and the extent to which observed mortality rates align with the study's predefined objectives. Furthermore, it underscores the challenges of detecting modest survival benefits in choroidal melanoma research and provides guidance for the design of future randomized trials in this field.

2. Materials and methods

2.1. Data collection

The raw data for this study were collected from the publicly available results in the original COMS publication, which detailed 12-year mortality rates after enucleation versus brachytherapy for patients with medium-sized tumors (Figure 1(B) of the original publication, published in 2006) [6]. The data included the number of patients at risk and the number of censored patients for each year up to 12 years for both the enucleation and brachytherapy arms. The number of events (deaths from melanoma) during each year after enrollment was calculated by subtracting the number of patients at risk in the current year from the number at risk in the previous year and further subtracting the number of censored patients during the interval. Specifically, the events were calculated using the equation:

$$\text{Events} = (\text{number at risk}_{\text{previous year}} - \text{number at risk}_{\text{current year}}) - \text{number censored}$$

Raw data were not requested from the COMS investigators, as the publicly available mortality rates were considered sufficient for the purposes of this threshold analysis. This study adhered to the tenets of the Declaration of Helsinki. As declared in the Swedish Ethical Review Act (2003:460), approval from the Swedish Ethical Review

Authority was not required as this study involved the re-analysis of publicly available data. It did not involve any treatments, tests, examinations, interviews, or other interventions with study participants. Additionally, it did not affect patients physically or psychologically, did not involve biological samples, and did not include any data or sensitive information that could be traced back to any individual.

2.2. Statistical analysis

The number of patients at risk, number of melanoma deaths, and number of censored patients were extracted from the COMS report for each year up to 12 years. We then conducted a threshold analysis to determine the number of additional deaths required in either the brachytherapy or enucleation arm to achieve a lower end of the 95% CI I for the risk ratio ≥ 1.00 . First, the standard error (SE) for the risk ratio at 12 years was calculated using the equation:

$$SE = \frac{\log(CI_{upper}) - \log(CI_{lower})}{2 \times 1.96}$$

Next, to determine the new risk ratio that corresponds to having a lower bound of the 95% CI ≥ 1.00 , the new log risk ratio (RR_{new}) was calculated using the equation:

$$\log(RR_{new}) = \log(1.00) + 1.96 \times SE$$

To adjust the number of deaths required to achieve this new risk ratio, we used the enucleation arm:

$$\text{Deaths}_{\text{new enucleation}} = \frac{\text{Deaths}_{\text{brachytherapy}} \times (N_2 - \text{censored}_{\text{enucleation}})}{RR_{new} \times (N_2 - \text{censored}_{\text{brachytherapy}})}$$

$$\text{Additional death}_{\text{enucleation}} = \text{Deaths}_{\text{new enucleation}} - \text{Deaths}_{\text{enucleation}}$$

And for the brachytherapy arm:

$$\text{Deaths}_{\text{new brachytherapy}} = \frac{\text{Deaths}_{\text{enucleation}} \times (N_1 - \text{censored}_{\text{brachytherapy}})}{RR_{new} \times (N_2 - \text{censored}_{\text{enucleation}})}$$

$$\text{Reduced deaths}_{\text{brachytherapy}} = \text{Deaths}_{\text{brachytherapy}} - \text{Deaths}_{\text{new brachytherapy}}$$

Where:

- N_1 is the total number of patients in the brachytherapy arm.
- N_2 is the total number of patients in the enucleation arm.
- $\text{Deaths}_{\text{brachytherapy}}$ and $\text{Deaths}_{\text{enucleation}}$ are the observed deaths in each arm.
- $\text{Censored}_{\text{brachytherapy}}$ and $\text{Censored}_{\text{enucleation}}$ are the numbers of censored patients in each arm.

In our initial analysis, we held the number of deaths constant in one arm while adjusting the other to achieve a statistically significant survival difference. This approach was subsequently expanded by presenting risk ratios as a function of changes in the number of deaths in either arm and by conducting a sensitivity analysis in which the number of deaths in each arm was uniformly adjusted by -40% , -20% , $+20\%$, and $+40\%$. All statistical analyses were conducted using R version 4.1.2, utilizing the tidy, pwr, and dplyr packages.

3. Results

First, we verified that our method, analyzing the data collected from the original publication, yielded results similar to those presented in the original publication. The data collected are detailed in [Table 1](#). At 12 years, there were 105 and 98 deaths with histopathologically confirmed melanoma metastasis in the brachytherapy

and enucleation arms, respectively. The risk ratio was 1.08 (95% CI 0.82 to 1.42), which is comparable to the published figures of 1.07 (95% CI 0.81 to 1.41).

3.1. Threshold analysis

To obtain a lower end of the 95% CI for the risk ratio greater than 1.00, it was found that 31 additional deaths would have been required in the brachytherapy arm, resulting in 136 deaths in the brachytherapy arm and 98 deaths in the enucleation arm, representing an increase of 30%. Alternatively, 23 fewer deaths in the enucleation arm, for the same number of deaths in the brachytherapy arm, would have favored enucleation, resulting in 75 deaths in the enucleation arm and 105 deaths in the brachytherapy arm, equivalent to a decrease of 23%.

Conversely, to make the risk ratio significantly higher for enucleation, 34 more deaths would be required in the enucleation arm, resulting in 132 deaths in the enucleation arm and 105 deaths in the brachytherapy arm, representing an increase of 35%. Alternatively, 27 fewer deaths in the brachytherapy arm would have favored

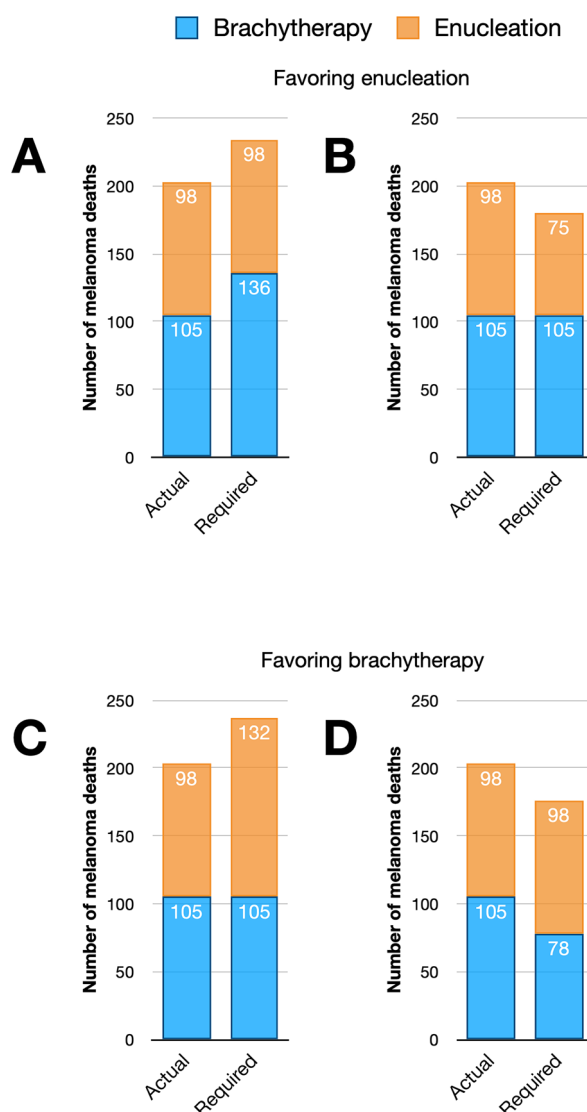


Figure 1. Threshold analysis of deaths with histopathologically confirmed melanoma metastasis required to favor enucleation or brachytherapy. Bars represent the additional or reduced deaths in each treatment arm required to demonstrate a statistically significant survival difference between brachytherapy and enucleation, defined as a 95% confidence interval (CI) excluding 1. (A) Adding 31 deaths due to histopathologically confirmed metastatic uveal melanoma in the brachytherapy arm would result in a risk ratio significantly favoring enucleation. (B) Reducing 23 deaths in the enucleation arm, while keeping the number of brachytherapy deaths constant, would also favor enucleation. (C) Conversely, adding 34 deaths in the enucleation arm would result in a risk ratio significantly favoring brachytherapy. (D) Reducing 27 deaths in the brachytherapy arm would similarly favor brachytherapy.

Table 1. Data collected from the original study.

Year after enrollment	Enucleation				Brachytherapy			
	<i>n</i> at risk	<i>n</i> censored	<i>n</i> melanoma deaths	<i>n</i> cumulative deaths	<i>n</i> at risk	<i>n</i> censored	<i>n</i> melanoma deaths	<i>n</i> cumulative deaths
0	660	0	0	0	657	0	0	0
1	655	2	3	3	646	7	4	4
2	640	9	6	9	627	11	8	12
3	602	12	26	35	593	18	16	28
4	572	17	13	48	555	18	20	48
5	533	19	20	68	532	11	12	60
6	476	45	12	80	481	41	10	70
7	414	56	6	86	429	46	6	76
8	354	54	6	92	366	53	10	86
9	298	53	3	95	305	53	8	94
10	246	50	2	97	250	51	4	98
11	191	54	1	98	196	49	5	103
12	135	56	0	98	140	54	2	105

Table 2. Required change in deaths for statistical significance by risk ratio.

Risk ratio	Change in deaths required for statistical significance (brachytherapy arm)
0.70	−10
0.75	−3
0.80	4
0.85	11
0.90	17
0.95	24
1.00	31
1.05	38
1.10	45
1.15	51
1.20	58
1.25	65
1.30	72

enucleation, resulting in 98 deaths in the enucleation arm and 78 deaths in the brachytherapy arm, corresponding to a decrease of 26% (Figure 1).

To explore the relationship between assumed treatment effects and the magnitude of additional deaths required, the analysis was expanded to include risk ratios ranging from 0.70 to 1.30 in increments of 0.05. The results, presented in Table 2, provide insights into the impact of varying risk ratios and statistical power on the required changes in mortality. For example, at a risk ratio of 0.70 (where values below 1 favor brachytherapy), 10 fewer deaths than observed would be sufficient to achieve statistical significance regardless of statistical power levels. As the risk ratio approaches 1 (indicating no treatment effect), the required additional deaths transition to positive values: at a risk ratio of 1.0, 31 additional deaths are needed. For risk ratios above 1 (favoring enucleation), the number of required additional deaths increases substantially. At a risk ratio of 1.25, 65 additional deaths are required, and at 1.30, 71.7 additional deaths are needed. Notably, the required number of additional deaths remains consistent across power levels for each risk ratio. Plotting the required changes in the number of deaths in each arm further illustrates the magnitude of the treatment effect that would have been required for statistical significance, and the interval of risk ratios from 0.60 to 1.05 where CIs overlap (Figure 2). This consistency reflects the fixed group sizes and observed mortality rates in the COMS dataset. These findings underscore the substantial survival differences required to achieve statistical significance, particularly when treatment effects are modest relative to the already high baseline mortality associated with metastatic uveal melanoma.

Negative values indicate the number of deaths that must be reduced (i.e., fewer events) in the brachytherapy arm to achieve statistical significance, while positive values indicate the number of additional deaths (i.e., more events) required.

3.2. Sensitivity analysis

To explore the effect of allowing mortality to vary proportionally in the two treatment arms, we adjusted the observed number of deaths by −40%, −20%, +20%, and +40% in each arm and recalculated the additional (or

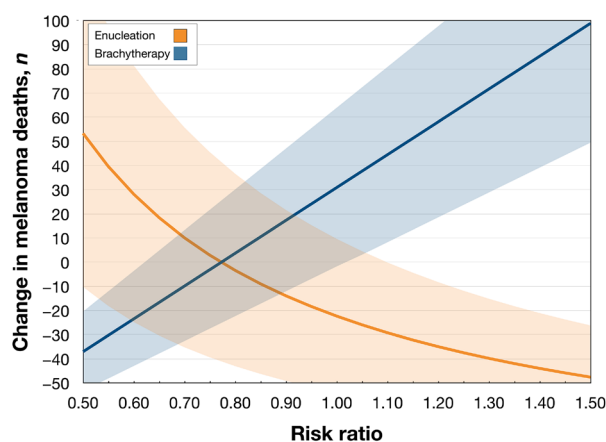


Figure 2. Threshold analysis of additional deaths required to achieve statistical significance for various risk ratios. The y-axis represents the change in melanoma-related deaths required in each treatment arm, while the x-axis represents the target risk ratio. The blue straight line corresponds to changes in the brachytherapy arm, and the orange curved line to changes in the enucleation arm. Shaded areas represent the 95% confidence intervals (CIs) for the additional deaths needed. Overlapping 95% CIs indicate nonsignificant differences between the risk ratios. Positive values on the y-axis indicate the number of deaths that would need to increase while negative values indicate the number of deaths that would need to decrease, to achieve the specified risk ratios.

Table 3. Sensitivity analysis: additional deaths required under percentage adjustments.

Target RR	Adjustment, %*	Direction**	Additional deaths [†]	95% CI lower	95% CI upper
0.70	−40	brachytherapy	−20	−30	−6
0.70	−40	enucleation	27	6	54
0.70	−20	brachytherapy	−26	−40	−8
0.70	−20	enucleation	36	8	72
0.70	+20	brachytherapy	−39	−60	−12
0.70	+20	enucleation	54	12	108
0.70	+40	brachytherapy	−46	−70	−14
0.70	+40	enucleation	63	14	126
1.30	−40	brachytherapy	17	−2	43
1.30	−40	enucleation	−13	−24	2
1.30	−20	brachytherapy	23	−3	57
1.30	−20	enucleation	−17	−32	3
1.30	+20	brachytherapy	35	−4	86
1.30	+20	enucleation	−25	−48	4
1.30	+40	brachytherapy	41	−5	100
1.30	+40	enucleation	−30	−56	5

CI: confidence interval; RR: risk ratio.

*The percentage by which the observed number of deaths in both treatment arms is uniformly adjusted.

**The treatment arm (brachytherapy or enucleation) for which the additional (or reduced) deaths required are calculated.

fewer) deaths required to obtain a 95% CI for the risk ratio that excludes 1.0. **Table 3** summarizes the results for target risk ratios of 0.70 (favoring brachytherapy) and 1.30 (favoring enucleation). These sensitivity results demonstrate that, even when the observed mortality is adjusted by up to $\pm 40\%$, the thresholds required for statistical significance remain high. For example, at a target risk ratio of 0.70, a -40% adjustment requires an additional reduction of 20 deaths in the brachytherapy arm (95% CI: -30 to -6) or an increase of 27 deaths in the enucleation arm (95% CI: 6 – 54). Similarly, at a target risk ratio of 1.30, a $+40\%$ adjustment necessitates an increase of 41 deaths in the brachytherapy arm (95% CI: -5 to 100) or a reduction of 30 deaths in the enucleation arm (95% CI: -56 to 5). These findings support our conclusion that the thresholds remain considerable under more flexible assumptions.

4. Discussion

Our study quantifies the additional or reduced deaths necessary to render the COMS trial results statistically significant. While the original trial concluded noninferiority between brachytherapy and enucleation, our threshold analysis elucidates the specific mortality changes required to demonstrate definitive superiority, thereby highlighting the trial's inherent statistical limitations. To achieve a 95% CI for the risk ratio excluding 1, the COMS trial

would have required a substantially larger difference in the number of deaths between treatment groups. This indicates that a survival benefit of considerable magnitude was necessary for the trial to demonstrate statistical significance in survival outcomes. Given that metastatic disease leading to death in the enrolled patients was likely already established before treatment, as indicated by previous studies, observing such a survival difference was perhaps unrealistic [8,9]. Even if some patients treated with brachytherapy had incompletely sterilized tumors, and surviving tumor cells in the eye led to additional seeding forming distant metastases, the impact of these cases would likely have been too small to achieve the required survival difference for statistical significance.

These findings should not be viewed as criticism of the COMS trial, which remains a landmark in ophthalmic oncology research. The study unequivocally demonstrated noninferiority in survival between iodine-125 plaque brachytherapy and enucleation for medium choroidal melanomas. However, it is essential to emphasize that the trial was designed with the goal of detecting superiority, not noninferiority. Specifically, the COMS trial's predefined goal of detecting a 25% minimum relative difference in 5-year mortality rates between the two treatment groups was not achieved [11].

In light of this, we argue that while the COMS trials for medium and large choroidal melanomas were statistically powered to assess equivalence or noninferiority in survival outcomes, they lacked sufficient power to conclusively demonstrate superiority.

To illustrate the implications of our findings for future trial design, assuming a baseline melanoma-specific mortality of approximately 15% at 12 years and aiming to detect a 25% relative increase (i.e., from 15% to 19%, corresponding to a risk ratio of 1.25), we estimate that—at a two-sided significance level of 0.05 and 80% power—approximately 1560 patients per treatment arm would be needed. This estimate represents more than a doubling of the original COMS sample sizes (approximately 660 per arm) and underscores the substantial recruitment efforts required to detect modest differences in survival outcomes in uveal melanoma. Alternatively, focusing on cohorts with a higher risk of mortality might improve statistical power. For instance, the disease-specific survival in the COMS trial was relatively favorable compared to other studies of uveal melanomas of similar size. One study reported a 10-year melanoma-related mortality of 28% in 677 tumors treated with plaque brachytherapy at a single Swedish institution, with a mean largest basal diameter of 10.1 mm (versus 11.4 mm in COMS) and a mean thickness of 5.2 mm (versus 4.8 mm in COMS), which raises questions regarding potential differences in the evaluation of deaths and cause-of-death determination [12,13]. Meticulous follow-up to capture all mortality events remains paramount.

Consequently, the observations from the COMS trial cannot be used to draw conclusions about superiority in overall or disease-specific survival outcomes between the two treatment modalities. Nonetheless, brachytherapy continues to serve as a principal local treatment option for appropriate tumors, most commonly employing iodine-125 or ruthenium-106 as radioisotopes.

In addition to iodine-125, ruthenium-106 is a well-established radioisotope used in plaque brachytherapy for uveal melanoma. Ruthenium-106 emits beta radiation with a shorter penetration depth and potentially causes less damage to surrounding healthy tissues, which is why it is often favored for thinner tumors, whereas iodine-125 emits gamma radiation with a less steep dose fall-off curve [14]. Both isotopes deliver localized radiation to the tumor while preserving the eye and potential vision, with comparable survival outcomes between the two. However, quality of life (QoL) is not necessarily better for patients treated with plaque brachytherapy than with enucleation [15]. In one QoL study within the COMS framework, patients treated with plaque brachytherapy reported better visual function initially, but this advantage diminished by 3 to 5 years post-treatment, and they were more likely to have symptoms of anxiety compared to patients treated with enucleation [16]. Conversely, later reports indicate that QoL can be worse in patients who undergo primary enucleation, although these findings may partly reflect older age, more advanced disease, and poorer prognoses among patients selected for enucleation, rather than an inherent consequence of the procedure itself [17].

4.1. Limitations

This study has several limitations. Firstly, this analysis bears resemblance to *post hoc* power analyses, which are limited by their reliance on observed *p* values. However, our approach focuses on determining thresholds for observed risk ratios, which provides a more robust framework for understanding statistical significance [18]. It is important to note that our threshold analysis emphasizes the magnitude of change required for the 95% CI to exclude 1.0, a conventional criterion for statistical significance. A modest difference in mortality—such as

the observed 7% relative increase (105 vs. 98 deaths) in the COMS trial—may nevertheless be clinically relevant. Although the trial was designed to detect a minimum relative difference of 25% in 5-year mortality rates and was not powered to capture small differences, these modest effects might still be important to patients and clinicians. Future research with larger cohorts, longer follow-up, or more sensitive endpoints is warranted to determine whether these small, undetected effects translate into meaningful clinical outcomes.

Second, our analysis also has limitations inherited from the original study. Using death with histopathologically confirmed melanoma metastasis as the outcome measure excludes patients who die with symptomatic or radiologically detectable metastases not verified by biopsy, leading to lower reported mortality than in other studies [19,20]. Given the challenges in determining the true cause of death, analyses of overall survival may be more reliable for the COMS study.

Lastly, our analysis relies on the simplifying assumption that deaths in one treatment arm remain constant while adjusting for deaths in the other. While this facilitates straightforward calculations, it does not reflect real-world follow-up patterns, where deaths would likely increase in both arms over time. Future analyses could incorporate proportional increases in deaths for both arms to provide a more realistic assessment.

Incorporating contemporary data on metastatic behavior could refine our threshold estimates. For instance, both older and recent studies indicate that a proportion of patients may present with undetected metastases at baseline, suggesting that adjusting for preexisting metastatic disease is essential for a more accurate assessment of treatment effects [9,21]. Additionally, it's essential to recognize that the COMS trial was conducted during a period with limited advancements in diagnostic and therapeutic modalities. Modern clinical trials benefit from enhanced imaging techniques, genomic profiling, and more sophisticated statistical methodologies, enabling more precise patient stratification and endpoint selection. Contemporary trials might prioritize endpoints like local control and quality of life, supported by biomarkers that better predict metastatic risk. These improvements underscore the evolving landscape of clinical trial design, which may limit the direct applicability of our threshold analysis to current research paradigms.

5. Conclusions

This hypothetical analysis demonstrates that achieving statistical significance in the COMS trial would have required a substantial survival benefit for either treatment, likely exceeding what could reasonably be expected given the metastatic behavior of uveal melanoma. These findings highlight the challenges of detecting modest treatment effects in randomized trials for uveal melanoma and underscore the importance of maintaining realistic expectations when interpreting trial results. Ensuring sufficient power to detect clinically meaningful differences, particularly in conditions with high baseline mortality, remains a pivotal challenge that necessitates continuous methodological advancements.

Author contributions

Viktor Gill: Writing- Reviewing and Editing, Validation. **Shiva Sabazade:** Writing- Reviewing and Editing, Validation. **Gustav Stålhammar:** Conceptualization, Methodology, Formal analysis, Investigation, Writing -Original Draft, Visualization, Supervision.

Disclosure statement

Gustav Stalhammar reports that financial support was provided by Region Stockholm. Gustav Stalhammar reports that financial support was provided by The Swedish Cancer Society. Gustav Stalhammar reports that financial support was provided by The Lormark family foundation. The authors have no other relevant affiliations or financial involvement with any organization or entity with a financial interest in or financial conflict with the subject matter or materials discussed in the manuscript apart from those disclosed.

No writing assistance was utilized in the production of this manuscript.

Ethical declaration

As declared in the Swedish Ethical Review Act (2003:460), approval from the Swedish Ethical Review Authority was not required as this study involved the re-analysis of publicly available data. It did not involve any treatments, tests,

examinations, interviews, or other interventions with study participants. Additionally, it did not affect the patients physically or psychologically, did not involve biological samples, and did not include collection or analysis of any data or sensitive information that could be traced back to any individual.

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Data availability statement

The raw data for this study were collected from the publicly available results in the original COMS publication, which detailed twelve-year mortality rates after enucleation versus brachytherapy for patients with medium-sized tumors (Figure 1B of the original publication including number at risk table) [6].

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**This report presents 12-year mortality data from the landmark COMS randomized trial comparing iodine-125 plaque brachytherapy and enucleation for medium-sized choroidal melanomas. It forms the foundational dataset for our threshold analysis,

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* This article provides estimates of tumor doubling times, suggesting that distant micrometastases may be established several years before the primary tumor is diagnosed.

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